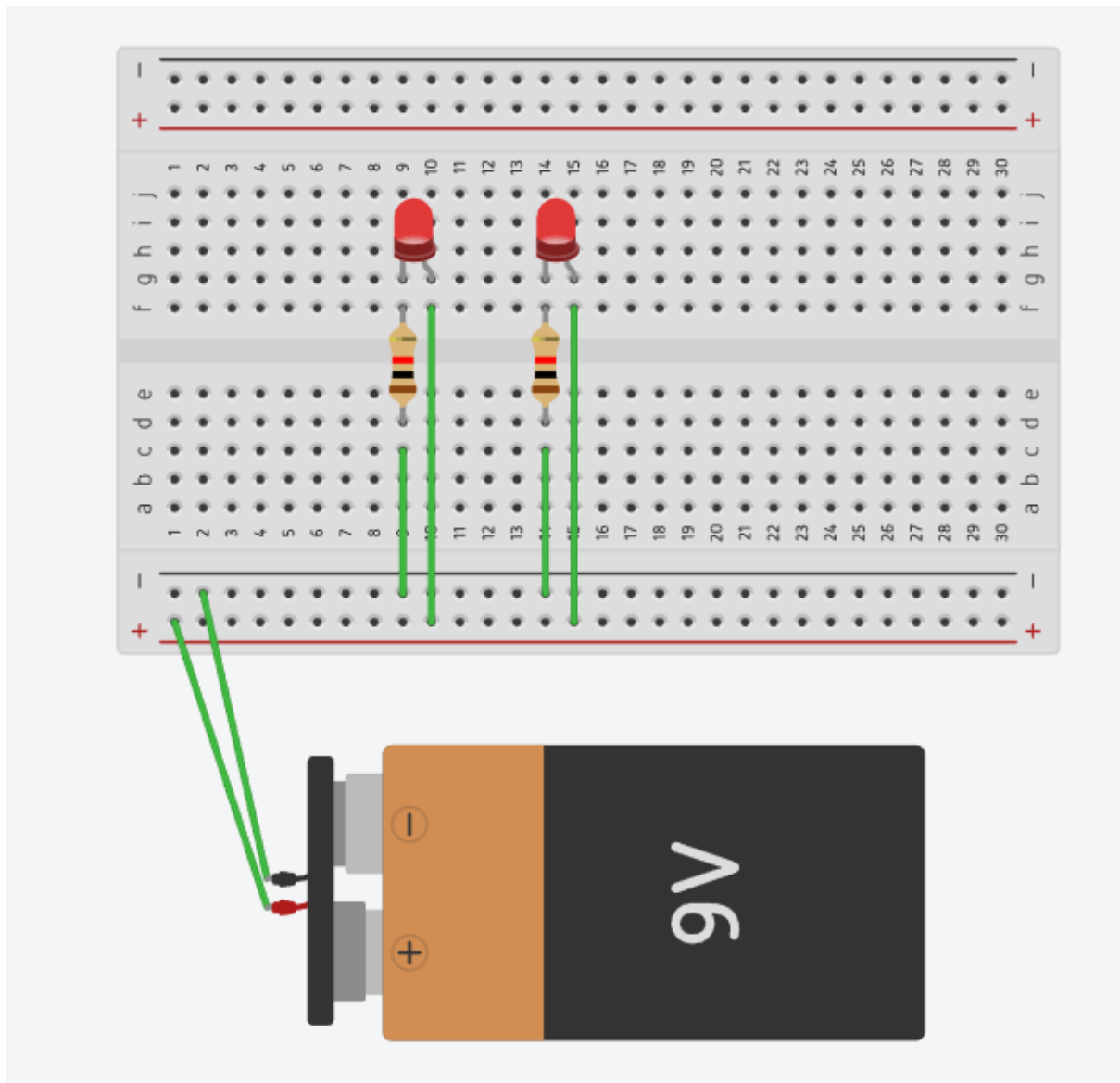




Description of Components

Breadboard

A breadboard is used to make temporary electronic circuits without soldering. It has many holes connected internally, which helps in easy connection of components.

LED (Light Emitting Diode)

An LED is a component that produces light when electric current flows through it. It has two legs:

- Long leg → positive (anode)
- Short leg → negative (cathode)

Resistor

A resistor controls the amount of current in the circuit. It is used to protect the LED from high current.

9V Battery

The 9V battery supplies electrical power to the circuit.
It has positive (+) and negative (–) terminals.

Connecting Wires

Connecting wires are used to connect different components on the breadboard and complete the circuit.

Description of Circuit Connection

The positive terminal of the 9V battery is connected to the positive rail of the breadboard.
The negative terminal of the battery is connected to the negative rail.

Each LED is connected in series with a resistor.

The resistor is connected to the positive rail, and the LED's negative leg is connected to the negative rail.

When the battery is connected, current flows from the positive terminal, through the resistor and LED, and returns to the negative terminal.

The resistor limits the current, so the LED glows safely.